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**NMAM INSTITUTE
OF TECHNOLOGY**

A MINI PROJECT REPORT

ON

**Smart IoT System for Detecting Current Leakage in Electric
Poles and Shock Prevention**

**Submitted in partial fulfillment requirements for the credit of
the Course on Internet Of Things (CC2002-1)**

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Abstract

Electric poles, whether made of metal or cement, pose a significant safety risk due to the potential for current leakage, which can lead to electric shocks for both humans and animals. This hazard can arise from faulty insulation, damaged wiring, or environmental conditions such as rain, where moisture increases the likelihood of electrical leakage. To address this issue, the proposed system presents a smart Internet of Things (IoT) solution designed to detect current leakage in electric poles, providing real-time alerts to prevent electric shocks. By integrating a combination of current and voltage sensors, an Arduino microcontroller, and an ultrasonic sensor, the system continuously monitors electrical conditions, activates alerts when hazardous situations are detected, and ensures safety by warning individuals who may approach the pole. This approach not only enhances the safety of public spaces but also minimizes the risk of electrocution in the vicinity of electric poles.

Introduction

Electric poles are integral components of modern power distribution networks, yet they also present an ongoing safety challenge, particularly when exposed to faulty insulation or environmental factors. The presence of current leakage from live wires—especially in wet conditions—can create serious electrical hazards that may go unnoticed without an automated detection system. Many of the current systems for detecting leakage are limited in their ability to provide real-time alerts or respond dynamically to changing conditions, making it difficult to protect people and animals from potential shocks. In this context, it is crucial to develop systems capable of detecting hazardous leakage before an accident occurs, especially in urban and rural environments where electric poles are prevalent.

Current research and solutions to this issue have mostly focused on either manual inspections or basic detection methods that do not provide immediate safety measures. While some IoT-based systems exist for monitoring electric infrastructure, most of these do not integrate real-time hazard alerts in response to leakage conditions or fail to account for proximity detection of individuals near dangerous poles. Therefore, there is a significant gap in developing a robust and real-time safety solution that can detect both current leakage and hazardous voltage levels, as well as determine if someone is in close proximity to a dangerous electric pole.

This report proposes a comprehensive solution to mitigate the risks of electric shock from current leakage in electric poles. Our system utilizes an IoT-based approach that combines current and voltage detection with proximity sensing to trigger alerts when hazardous conditions are present. Through the use of an Arduino microcontroller, ultrasonic sensors, and a voltage threshold-based alert mechanism, the system provides a reliable and efficient way to monitor electric poles. The integration of a fail-safe design ensures that only genuine hazards activate the alert system, reducing the occurrence of false alarms and increasing safety for nearby individuals. This work aims to improve the safety of electrical infrastructure, preventing shocks and potential fatalities caused by undetected current leakage.

Problem Statement:

Electric poles, whether made of metal or cement, pose a significant safety risk due to the potential for current leakage, which can occur even when there are no visible signs of damage. Faulty insulation, worn-out wiring, or moisture intrusion during rainstorms can cause live electrical wires to leak current, potentially energizing the pole's surface. This creates a hazardous situation for anyone who comes into contact with the pole, including pedestrians, workers, and animals, particularly in areas where poles are exposed to fluctuating weather conditions. Even cement poles, which are less conductive than metal, can carry electrical current when saturated with water or exposed to prolonged moisture.

Often, these leaks are not easily detected through standard inspection methods, leaving poles vulnerable to causing electric shocks. To address this issue, an automated system is needed that can continuously monitor and detect current leakage in real time. Using sensors or alarms, the system could warn people in close proximity through audible or visual signals alerting them to the potential danger of electric shock.

Literature Survey

4.1 Electrocutation Prevention Using IoT

In this paper[1] "Electrocutation Prevention Using IoT" proposes an IoT-based system to enhance electrical safety by detecting and alerting users to potential electrical hazards in real-time, making it valuable for both residential and industrial environments. The approach includes monitoring key electrical parameters with sensors, using a microcontroller for data processing, and establishing cloud connectivity and mobile notifications for immediate user alerts. Results show the system's effectiveness in anomaly detection, potentially reducing electrocution risks through real-time monitoring, though it faces challenges such as reliance on continuous power, cybersecurity risks, and setup costs. The study highlights IoT's role in advancing electrical safety despite these limitations.

4.2 IoT-Based Electric Pole Safety System

In this paper[2] "IoT-Based Electric Pole Safety System" addresses the urgent issue of electrocution accidents caused by faulty electric poles. It presents an IoT solution that uses sensors and a microcontroller to monitor and detect faults, notifying authorities and disconnecting power automatically to prevent accidents. The system includes real-time fault detection and automated notifications via an LCD display and a voice module for local alerts. Hardware components such as the ATmega328 microcontroller, voltage and current sensors, and NodeMCU (ESP8266) enable fault detection and communication. While results show improved safety and rapid response, the system requires stable internet, skilled technicians for installation, and environmental adjustments for sensor accuracy. Future developments may incorporate biometric security and mobile applications for expanded functionality.

4.3 IoT Based SLT (Smart Line Transmission) Pole with Electrical Lineman Safety

In this paper[3] "IoT-Based SLT Pole with Electrical Lineman Safety" addresses the issue of electrical accidents from broken power lines by introducing an IoT-based system that detects line breaks and alerts authorities while also preventing unauthorized access. The system uses RF transceivers to monitor line integrity, providing automatic

alerts through SMS notifications with GPS location when faults occur. Additionally, a password-protected circuit breaker allows only authorized personnel to manage power lines. Key components include an Arduino microcontroller, RF transceivers, GPS and GSM modules, a relay, and a keypad. This solution enhances public and lineman safety, though it requires reliable wireless connectivity and skilled technicians for installation. Future improvements may involve mobile applications and biometric security.

4.4 Leakage Current Detector and Warning System Integrated with Electric Meter

In this paper[4] introduces a new power converter design tailored for renewable energy systems to improve efficiency and reliability. Traditional converters often lose energy due to fluctuating energy inputs, so the authors propose a system that uses advanced control techniques, such as model predictive control, for optimal real-time energy management. The system includes power semiconductors, microcontrollers, batteries, and sensors, tested through simulations and experiments. Results indicate a significant efficiency boost over conventional designs, making it suitable for renewable applications, though it adds complexity and cost. Overall, the study offers a promising approach to enhance energy conversion in renewable technologies.

4.5 Research on Electric Shock Prevention and First Aid for Electric Vehicles

In this paper[5] addresses the risk of electric shock in electric vehicles (EVs), which operate at high voltages between 200-600V, posing safety concerns. The authors propose a comprehensive safety framework that includes both direct and indirect protection strategies, emphasizing preventive measures and first aid protocols. Key techniques include remote monitoring platforms for safety tracking, insulation fault monitoring, and high-voltage training for maintenance personnel. Results highlight the need for these measures and proper training to minimize risks. While enhancing safety, these strategies may add complexity and costs. Overall, the paper underscores essential safety protocols to prevent electric shocks in EVs.

4.6 Introduction to Electric Shock Protection

In this paper[6] explores electric shock hazards in electrical installations and examines protection systems primarily based on UK regulations but adaptable worldwide. It identifies electric shock as a serious safety concern, often occurring when the human body becomes part of an electrical circuit. Key protection mechanisms discussed include earthing, equipotential bonding, and Residual Current Devices (RCDs), which disconnect the power in fault conditions. The study provides a descriptive analysis of factors affecting electric shock severity, such as current and duration, and outlines various earthing systems (e.g., TN, TT). Although focused on UK standards, the paper emphasizes the global importance of timely disconnection (within 0.4 seconds) to reduce risks, with RCDs recommended when other measures may be insufficient.

4.7 An Ultrasonic Sensor for Distance Measurement in Automotive Applications

In this paper[7] introduces a low-cost ultrasonic sensor designed to measure vehicle height from the ground, supporting automotive applications like parking assistance and headlight leveling. Using a 40 kHz piezoelectric transducer, it calculates distance via time-of-flight measurements, adjusting for noise and temperature variations to improve accuracy. The system features constrained optimization for precise echo detection, automatic gain control, and temperature compensation, achieving a 1 mm accuracy within speeds up to 30 m/s. Tests show it performs well under moderate noise and temperature ranges, though accuracy declines in high-speed or extreme temperature conditions. This sensor offers an affordable, adaptable solution for vehicle height measurement, with potential limitations in extreme settings.

4.8 Review of Ultrasonic Ranging Methods and Their Current Challenges

In this paper[8] analyzes ultrasonic ranging methods, highlighting their wide-ranging applications in fields such as automotive, industrial automation, and healthcare. While ultrasonic sensors are effective for accurate distance measurement, they face challenges like environmental noise, multipath interference, and temperature sensitivity.

The authors examine various sensing techniques, including time-of-flight, phase shift, and frequency modulation, as well as improvements like noise-canceling and adaptive filtering to enhance accuracy. Despite advancements, issues with long-range accuracy and robustness remain, prompting a call for further research in algorithm refinement and sensor fusion (e.g., with radar or LiDAR). The paper underscores ultrasonic sensors' adaptability but notes limitations in noisy or extreme environments, suggesting that sensor fusion and advanced processing may boost future performance.

4.9 Calibration of ZMPT101B Voltage Sensor Module Using Polynomial Regression for Accurate Load Monitoring

In this paper[9] focuses on improving the accuracy of the ZMPT101B voltage sensor, commonly used for AC voltage measurement in smart meters and load monitoring. The researchers employed polynomial regression, specifically using a third-order polynomial, to calibrate the sensor and reduce errors in voltage readings. By correlating the sensor's ADC output with the actual input voltage, the method significantly improved measurement accuracy, reducing errors to less than 1% for peak-to-peak calculations. This approach, while cost-effective and suitable for high-accuracy applications like smart grids, requires careful calibration and may have limitations in adapting to voltage ranges outside the calibrated model.

Proposed System

The proposed system is designed to detect current leakage in electric poles (represented practically in our model by a conductor) and activate an alert system only when both current leakage and high voltage levels are detected. The system functions as follows:

1. Current and Voltage Detection:

- A wire will be wrapped around the circumference of the conductor (pole).
- This wire will be connected to a voltmeter, which detects the voltage across the conductor.
- The sensed voltage values will be transmitted to an Arduino for processing.

2. Threshold-Based Safety Mechanism:

- The Arduino will continuously monitor the voltage values detected from the conductor.
- If the current is detected and the detected voltage exceeds the predefined threshold, it indicates potential current leakage.
- Only when both conditions are met, the system activates the alert mechanism.

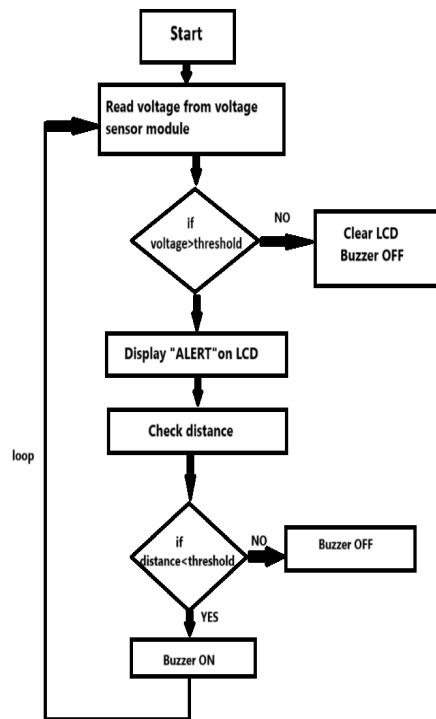
3. Alert System:

- **Ultrasonic Sensor:** An ultrasonic sensor will detect if anyone is approaching the conductor. This adds an additional layer of safety by ensuring that the alert system is relevant only when a person is nearby.
- **Buzzer:** Once current leakage is detected and voltage surpasses the threshold, a buzzer will be activated to warn the nearby individuals about the danger.
- **LCD Display:** Simultaneously, an LCD display will show a warning message indicating it is unsafe to approach the pole.

4. Fail-Safe Design:

- If voltage does not exceed the threshold, the alert system remains inactive, preventing false alarms and ensuring alerts are only triggered under hazardous conditions.

Methodology



Flowchart of entire process

The flowchart represents a step-by-step methodology for a system that monitors voltage and distance, triggers alerts, and activates a buzzer based on threshold conditions. Here's a detailed breakdown:

1. Start: The process begins.
2. Read voltage from the sensor: The system reads the voltage value from a connected sensor.
3. Check if voltage > threshold:
If the voltage is greater than a predefined threshold:
YES: Continue to the next step.
NO: Skip to "Clear LCD, Buzzer OFF."

4. Display "ALERT" on LCD: If the voltage exceeds the threshold, the system displays an "ALERT" message on an LCD screen.

5. Check distance: The system then checks the distance value from another sensor (e.g., an ultrasonic sensor).

6. Check if distance < threshold:

If the distance is below a specific threshold:

YES: The buzzer is turned ON, alerting about the proximity issue.

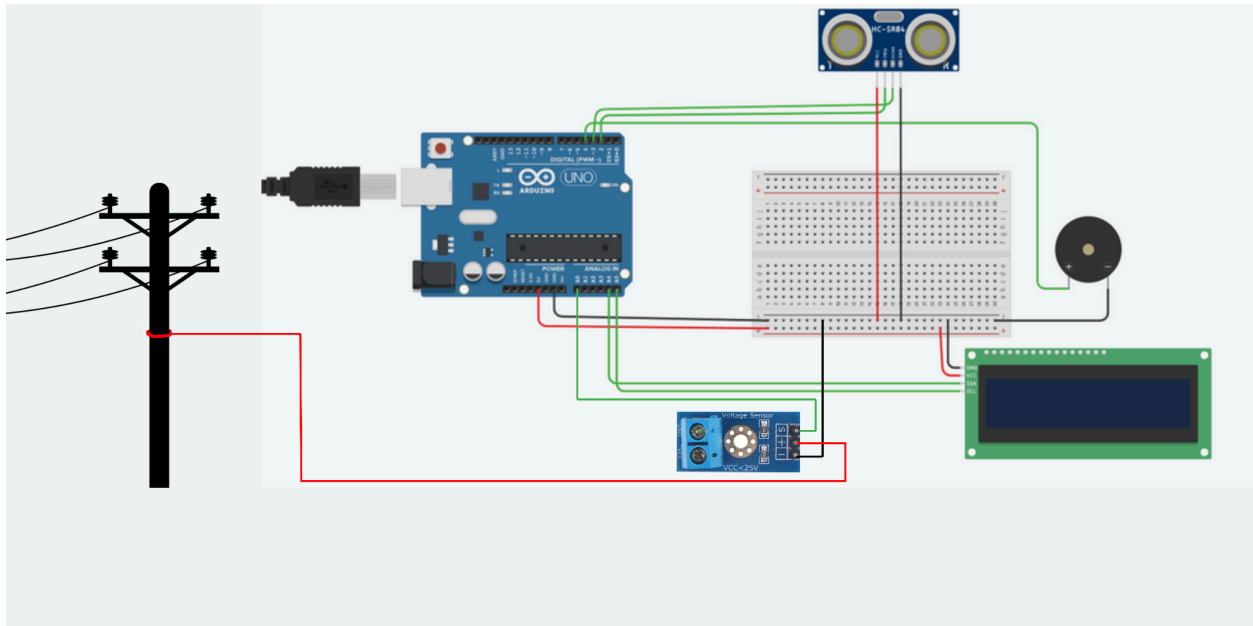
NO: The buzzer remains OFF.

7. Loop: The process goes back to the beginning to continuously monitor voltage and distance.

8. Clear LCD, Buzzer OFF: If the voltage is below the threshold, the system clears the LCD screen and turns off the buzzer.

This methodology is intended to provide real-time alerts based on specific voltage and distance conditions, ensuring a quick response when necessary.

Circuit Diagram



Circuit design

Description

The circuit consists of an Arduino Uno connected to a voltage sensor, ultrasonic sensor, buzzer, and LCD display. The voltage sensor detects potential leakage, while the ultrasonic sensor ensures activation of the alert system only when a person is nearby. The Arduino processes the data and triggers the buzzer and LCD alert if both current leakage and high voltage are detected.

Software and Hardware requirements

Software requirements

Arduino IDE: The Arduino IDE (Integrated Development Environment) is a popular open-source software platform used for programming and uploading code to Arduino boards. It provides a simple, user-friendly interface for writing, compiling, and debugging code in a language similar to C/C++. The IDE includes a built-in text editor, where users can write code, and an output console to display error messages or compile information. Arduino IDE also comes with a library manager, allowing users to access and include various libraries to expand the board's functionality, like handling sensors, displays, and communication modules. It supports code uploading directly to the board via USB, making it easy for beginners and experienced developers alike to bring their projects to life. The IDE is compatible with Windows, macOS, and Linux, and is widely used in educational, DIY, and prototyping environments for building projects that range from simple LED blinkers to complex IoT devices.

Hardware requirements

Arduino: Arduino is an open-source electronics platform based on easy-to-use hardware and software. It consists of microcontroller boards, such as the Arduino Uno, that can read inputs from various sensors and control outputs like LEDs, motors, or displays. The platform is highly accessible for beginners and versatile enough for advanced users, allowing the creation of interactive projects through programming in the Arduino IDE. Known for its wide range of applications, Arduino is popular in prototyping, education, and DIY electronics, enabling creators to build everything from simple gadgets to complex systems.

Buzzer: A buzzer is a small audio signaling device that emits sound when powered. In Arduino projects, it's commonly used to provide audio feedback, alerts, or sound effects, typically operating at varying frequencies to create different tones.

LCD I2C: This Liquid Crystal Display module uses I2C (Inter-Integrated Circuit) communication, which reduces the number of pins needed to connect it to an Arduino. It's ideal for displaying data, like text or numbers, and is easy to integrate, thanks to its efficient two-wire interface.

Breadboard: A breadboard is a tool for prototyping electronic circuits without soldering. It consists of interconnected sockets where you can insert components and jumper wires, allowing for quick adjustments and testing of Arduino circuits.

Ultrasonic Sensor: An ultrasonic sensor measures the distance to objects by emitting sound waves and calculating the time it takes for the echo to return. Often used in obstacle detection, it's popular in Arduino projects like distance measurement and robotics.

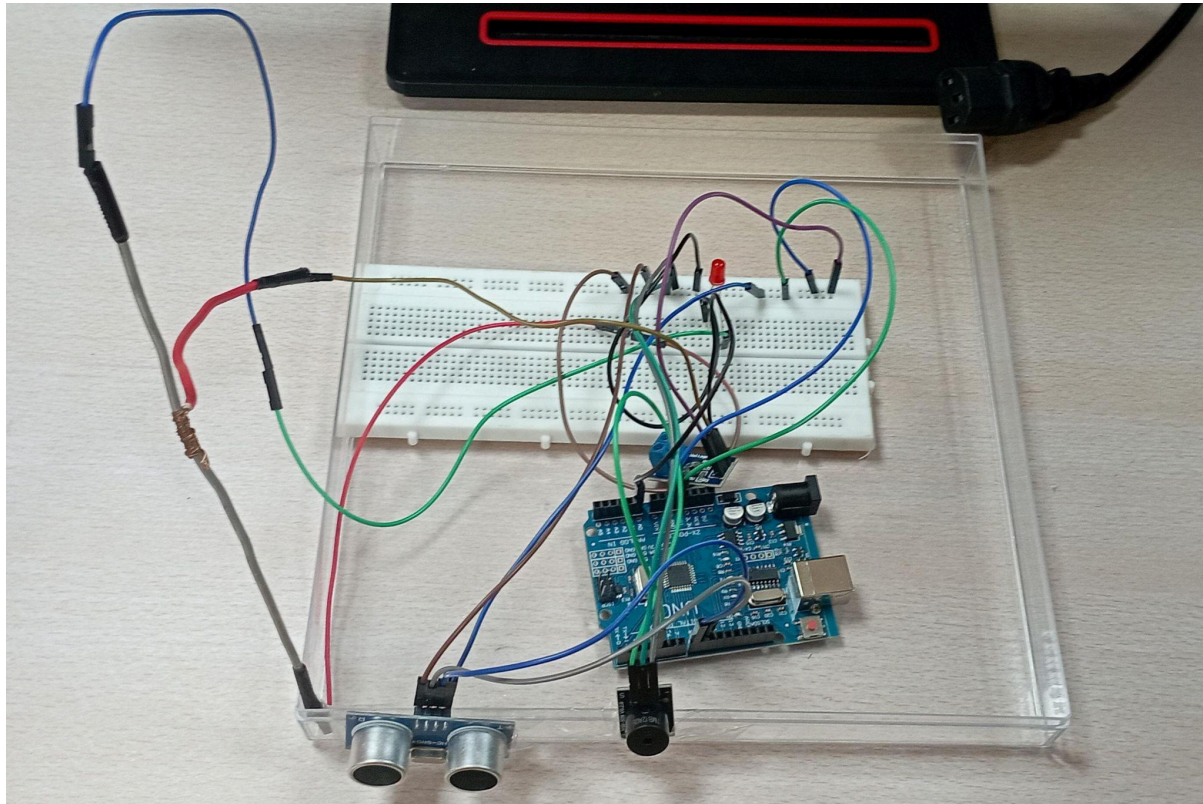
LED: Light Emitting Diodes (LEDs) are small lights used in Arduino projects to indicate status or display visual feedback. They're easy to control, and with PWM (Pulse Width Modulation), you can vary their brightness to create effects.

Jumper Wires: These are flexible wires with pin connectors used to link components on a breadboard or connect them to an Arduino. They come in male-to-male, male-to-female, and female-to-female types, making prototyping easy and versatile.

USB Cable: A USB cable is essential for connecting the Arduino board to a computer. It allows you to upload code and also powers the board, facilitating quick development and debugging during project creation.

Voltage Sensor Module: This module is designed to measure the voltage of connected circuits, ideal for monitoring battery levels or external power supplies in Arduino projects. It's especially useful for battery-operated or energy-efficient projects.

Result and Discussion



System for Detecting Current Leakage in Electric Poles and Shock Prevention

The Smart IoT System for Detecting Current Leakage in Electric Poles and Shock Prevention was successfully implemented, demonstrating its ability to monitor, detect, and alert in cases of potential electric shock hazards. The system effectively achieved the following outcomes:

1. Real-Time Voltage and Current Monitoring: The Arduino-based system continuously monitored the voltage across the conductor, simulating the electric pole. This real-time monitoring enabled quick detection of abnormal voltage levels due to current leakage.

2. Threshold-Based Safety Activation: The system activated the alert mechanism only when both conditions were met: detection of current leakage and a voltage level above the predefined threshold. This selective activation prevented unnecessary alerts and ensured the system was only active in hazardous situations.

3. Proximity Detection with Ultrasonic Sensor: The ultrasonic sensor successfully detected individuals approaching the electric pole (or conductor). This additional safety layer prevented unnecessary alerts, activating the warning only when someone was in close proximity to the danger zone.

4. Effective Alert Mechanisms: Upon meeting the hazardous conditions, the buzzer sounded, and the LCD display showed a clear warning message. These alerts were timely and effective in communicating the potential danger to people nearby, demonstrating the system's capacity for real-time intervention.

Conclusion and Future Enhancements

The Smart IoT System for Detecting Current Leakage in Electric Poles and Shock Prevention successfully enhances public safety by detecting current leakage and activating alerts only when hazardous conditions are met. Through real-time monitoring, threshold-based activation, and proximity detection, the system minimizes false alarms and effectively warns individuals of potential electric shock risks. This adaptable, efficient solution demonstrates how IoT technology can mitigate urban electrical hazards, with potential for further improvements and large-scale deployment in smart city infrastructure.

Future enhancements could include the integration of machine learning algorithms to predict and prevent potential electrical faults based on historical data, improving the system's ability to anticipate and address risks before they occur. Additionally, incorporating wireless energy harvesting technology could help make the system more energy-efficient and self-sustaining. Advanced data analytics and cloud-based monitoring platforms could also enable predictive maintenance and real-time decision-making, further optimizing system performance. Integration with mobile applications could offer users more personalized notifications and enhance user interaction with the system, ensuring faster response times in emergency situations. Lastly, expanding the system's scalability to cover a wider range of infrastructure, such as residential buildings and industrial complexes, could maximize its impact on urban safety.

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